Pratibimb architecture

A digital twin for hypertension. Two streams in, a personal model of the body in the middle, a calibrated 72 hour warning and a consented check-in ladder out.

the record also feeds the fusion model's record tower

Two streams in

Synthetic only (sandbox rule)

Health record (static)

Synthea-shaped FHIR R4 bundles
Conditions (SNOMED), labs (LOINC)
Medicines, adherence, family history
Polygenic risk tier, Framingham inputs

Wearable stream (dynamic)

Nightly resting HR, HRV, sleep, steps
SpO2, respiratory rate
Home BP cuff readings
5 minute intraday trace, live stream

Device sync (companion app)

Health Connect on Android, HealthKit on iOS
Oura and Whoop webhooks, CSV upload
POST /api/ingest with a freshness chip

Synthetic cohort generator

240 patients x 120 days, seed 42
Labelled strain episodes (3 kinds)
Colds, missed doses, travel, late nights
Hidden true response slopes per person

Personal Physiology Engine

The twin: how this body responds

1 Personal baseline

28 day resting HR median
60 day HRV mean and smallest worthwhile change
7 day home systolic slope, backward windows only

2 Personal response slopes

Sleep to heart rate, sleep to HRV
Steps to heart rate, steps to blood pressure
Calm days only, AR(1)-corrected errors

3 Empirical Bayes pooling

DerSimonian and Laird partial pooling
New users borrow from the cohort
Recovers the hidden truth: r 0.59 to 0.71

4 Twin-adjusted signals

Unexplained HR, HRV and BP change
Record-aware reading (beta blockers, diabetes)
Lifestyle load: 7 day sleep debt, activity
Expanding-window slopes: no future leaks in

Prediction

Validated on patients the model never saw

Fusion model

LightGBM on watch, cuff and twin signals
Eight-feature record tower
Patient-level CV, 3 splits x 5 folds

Calibration and band

Platt scaling on out-of-fold scores
P10 to P90 band from five fold models

Operating points

Urgent: most sensitive within 0.045 false alerts per patient-day. Watch: 90% of episodes

Forecasts

18 quantile models: HR and BP, days 1 to 3

Explanation

SHAP grouped into record and watch
NightSignal-style rule runs alongside

Action layer

FastAPI, SQLite, live stream (SSE)

- **Alerts and budget meter**
  Acknowledge, snooze, escalate
- **Thresholds, 14 day replay**
  Preview before you apply
- **What-if simulator**
  Uses this person's own slopes
- **Check-in ladder**
  Patient, 2 contacts, doctor, 112
- **Consent and audit log**
  Append-only, consent basis
- **FHIR and ABDM export**
  WellnessRecord bundle

People

Who acts

Doctor

Triage list
Twin page
Simulate
Alerts, model card

Patient

English and Hindi
What helps you
Check-in
Press-and-hold SOS

Contacts

SMS, WhatsApp and voice (mocked, DLT header shown)

Results on the synthetic cohort (mean of three patient splits)

0.876	0.851	70%	48 h
AUROC, fused twin	AUROC without the twin	episodes caught at 0.045 false alerts per patient-day	median warning before the event

Trust built in

- **Synthetic data only** — seeded and reproducible
- **DPDP-ready consent** — per stream, withdraw and delete
- **No AI prescribing** — patient text never changes doses
- **CDSCO SaMD path** — likely Class B or C
- **Leakage tests** — no future values, patient-level split